

Tutorial 6: AI-Assisted Risk Modelling and Diversification in R

(TEACHER'S VERSION WITH EXPANDED SOLUTIONS)

Theme: From Risk Register to Correlation, Simulation and Portfolio Decisions

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

Synergy & Comparison

The Logic of this Tutorial: Lecture 6 brought us from qualitative risk management to quantitative probability, covariance, correlation and portfolio diversification. Tutorial 5 showed how R can help us analyse stocks, bonds, volatility, drawdowns and portfolio weights. Tutorial 6 connects both ideas. We begin with a risk register, turn it into simple data, model frequency and severity, simulate uncertainty, measure co-movement, and finally explore how correlation changes the shape of the efficient frontier. The central message is that a future actuary should not merely calculate numbers; they should understand how uncertainty enters decisions, how risks interact, and when diversification really works.

D. Dependence: Covariance, Correlation and the Hidden Link Between Risks

Key Definition / Result

For two random variables X and Y ,

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

The correlation coefficient is

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho_{X,Y} \leq 1.$$

Covariance and correlation tell us whether two uncertain quantities tend to move together, move in opposite directions, or have little linear co-movement.

Synergy & Comparison

Why this matters: In Tutorial 5, students observed that combining SPY and AGG could reduce portfolio risk. Lecture 6 explained the mathematical engine: the variance of a linear combination includes a covariance term. Tutorial 6 now lets students manipulate correlation directly and see how it changes risk.

Suggested AI Prompt: Simulating Correlated Returns *** Advanced

"Act as a quantitative finance tutor. Please write beginner-friendly R code to simulate two asset returns with a chosen correlation (e.g., -0.5). Since I only know the Uniform distribution, use runif() to generate the base returns and independent noise. Then calculate the sample correlation and plot the points."

Investigation / Activity: Creating Two Artificial Assets With Chosen Correlation *** Advanced

Concept Scaffold: Instead of relying on real market data, we create artificial return data using the Uniform distribution. By carefully mixing a shared variable and an independent variable, we can control the correlation!

Teaching Note / Code Solution:

```
set.seed(2026)

number_of_days <- 1000
chosen_correlation <- -0.5

# Step 1: Create a base sequence of uniform returns for Asset A.
# We center it at 0 so our correlation mixing math works cleanly.
asset_A_centered <- runif(
  n = number_of_days,
  min = -0.02,
  max = 0.02
```

```

)

# Step 2: Create independent noise using the same uniform spread.
independent_noise <- runif(
  n = number_of_days,
  min = -0.02,
  max = 0.02
)

# Step 3: Create Asset B by mixing Asset A and the noise.
# This mathematical trick guarantees the correlation we want!
asset_B_centered <- chosen_correlation * asset_A_centered +
  sqrt(1 - chosen_correlation^2) * independent_noise

# Step 4: Add a small average expected return to both (e.g., 0.05% daily
  )
asset_A_return <- asset_A_centered + 0.0005
asset_B_return <- asset_B_centered + 0.0005

# Check the generated correlation
cor(asset_A_return, asset_B_return)

plot(
  asset_A_return,
  asset_B_return,
  pch = 16,
  col = rgb(0, 0, 1, 0.35),
  main = "Simulated Asset Returns with Chosen Correlation",
  xlab = "Asset A Daily Return",
  ylab = "Asset B Daily Return"
)
abline(h = 0.0005, v = 0.0005, col = "gray")

```

Teacher Note: The linear combination formula $\rho X + \sqrt{1 - \rho^2}Y$ works to set the exact correlation ρ for *any* two independent variables X and Y as long as they have mean 0 and equal variance. Because we strictly used `runif(-0.02, 0.02)` for both, the math holds perfectly without needing Normal distributions! Students should see that when correlation is negative, the scatter plot tends to slope downward. This cre-

ates a potential diversification benefit.

Reflection Question: Correlation as a Story

In words, what does $\rho = -0.5$ mean? Does it mean that one asset always goes up when the other goes down? Why or why not?

Common Mistake

Correlation is not the same as causation. If two assets move together, it does not automatically mean one causes the other to move. There may be a shared economic factor, such as interest rates, inflation expectations or market sentiment.

E. The Diversification Engine: Portfolio Variance

Key Definition / Result

For a two-asset portfolio with weights w_A and $w_B = 1 - w_A$, expected return is

$$E[R_p] = w_A E[R_A] + w_B E[R_B].$$

Portfolio variance is

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho_{AB} \sigma_A \sigma_B.$$

The final term,

$$2w_A w_B \rho_{AB} \sigma_A \sigma_B,$$

is the **dependence term**. It is the mathematical reason why diversification may work or fail.

Suggested AI Prompt: Manual Portfolio Risk Function ★★ Intermediate

"Please help me write an R function that calculates the expected return and volatility of a two-asset portfolio. The function should take asset weights, ex-

pected returns, volatilities and correlation as inputs. Use clear variable names suitable for beginners."

Worked Example: Portfolio Mean and Volatility Function

★★

Intermediate

Teaching Note / Code Solution:

```
rm(list = ls())

calculate_portfolio_stats <- function(weight_asset1, weight_asset2,
                                     return_asset1, return_asset2,
                                     volatility_asset1, volatility_asset2,
                                     correlation) {

  if (abs(weight_asset1 + weight_asset2 - 1) > 1e-6) {
    warning("Heads up: Your asset weights do not sum to 1 (100%).")
  }

  # Step 1: Expected Return
  expected_portfolio_return <- (weight_asset1 * return_asset1) + (weight_
    asset2 * return_asset2)

  # Step 2: Variance
  variance_asset1 <- volatility_asset1^2
  variance_asset2 <- volatility_asset2^2
  covariance <- volatility_asset1 * volatility_asset2 * correlation

  portfolio_variance <- (weight_asset1^2 * variance_asset1) +
    (weight_asset2^2 * variance_asset2) +
    (2 * weight_asset1 * weight_asset2 * covariance)

  # Step 3: Volatility
  portfolio_volatility <- sqrt(portfolio_variance)

  # Step 4: Output
  results <- list(
    Expected_Return = expected_portfolio_return,
    Volatility = portfolio_volatility
```

```
)  
  
  return(results)  
}  
  
my_portfolio <- calculate_portfolio_stats(  
  weight_asset1 = 0.60,  
  weight_asset2 = 0.40,  
  return_asset1 = 0.10,  
  return_asset2 = 0.05,  
  volatility_asset1 = 0.15,  
  volatility_asset2 = 0.05,  
  correlation = 0.20  
)  
  
print(my_portfolio)
```

The function makes the formula operational. Students should identify that changing only the correlation can change the portfolio volatility even when the asset volatilities and expected returns remain fixed.

Investigation / Activity: Changing Correlation While Keeping Everything Else Fixed ★★

Intermediate

Task. Compare portfolio volatility when correlation is +1, 0, -0.3, and -1.

Teaching Note / Code Solution:

```
# 1. Define the different correlation values we want to test  
correlation_values <- c(1, 0, -0.3, -1)  
  
# 2. Create an empty vector to store the volatilities we calculate  
portfolio_volatilities <- c()  
  
# 3. Use a 'for' loop to run our calculation for EACH correlation value  
for (cor in correlation_values) {  
  
  # Call the function we created earlier
```

```
temp_portfolio <- calculate_portfolio_stats(  
  weight_asset1 = 0.60,  
  weight_asset2 = 0.40,  
  return_asset1 = 0.10,  
  return_asset2 = 0.05,  
  volatility_asset1 = 0.15,  
  volatility_asset2 = 0.05,  
  correlation = cor # <--- We plug in the current correlation from our  
    loop here!  
)  
  
# Extract the Volatility from the result and add it to our storage  
  vector  
portfolio_volatilities <- c(portfolio_volatilities, temp_portfolio$  
  Volatility)  
}  
  
# 4. Combine the correlations and their resulting volatilities into a  
  neat table  
comparison_table <- data.frame(  
  Correlation = correlation_values,  
  Volatility = portfolio_volatilities  
)  
  
# 5. Print the final comparison table  
print(comparison_table)
```

As correlation decreases, portfolio volatility decreases. The expected return does not change because expected return depends on the weighted average of returns, while volatility depends on variances and covariance. This distinction is central to portfolio thinking.

Reflection Question: The Deep Idea

Why can correlation change portfolio risk without changing portfolio expected return? What does this tell us about the difference between return and risk?

F. Exploring the Efficient Frontier by R

Synergy & Comparison

Extension of Tutorial 5: Tutorial 5 used real assets, SPY and AGG, to explore a minimum-variance portfolio and the Sharpe ratio. Here we build a cleaner theoretical experiment. We choose two artificial assets and sweep the weight from 0% to 100%. Then we repeat the experiment under different correlations to see how the efficient frontier bends.

Suggested AI Prompt: Efficient Frontier Under Different Correlations

*** Advanced

"Act as an R programming mentor. I want to explore a two-asset efficient frontier. Asset A has expected return 12% and volatility 20%. Asset B has expected return 8% and volatility 10%. Test portfolio weights from 0% to 100% in 1% steps. Plot expected return against volatility. Repeat this for correlations +1, 0, -0.3 and -1. Use clear names and beginner-friendly comments."

Investigation / Activity: Drawing Several Frontier Curves*** Advanced

Teaching Note / Code Solution:

```
# If you don't have ggplot2 installed, you will need to run this line
# once:
# install.packages("ggplot2")

# Load the library so we can use its plotting functions
library(ggplot2)

# 1. Define our asset characteristics (using decimals)
return_asset_A <- 0.12
volatility_asset_A <- 0.20

return_asset_B <- 0.08
volatility_asset_B <- 0.10
```



```

# 2. Define the correlations and the sequence of weights to test
correlations_to_test <- c(1, 0, -0.3, -1)

# seq() creates a sequence of numbers. Here, from 0 to 1 in steps of
  0.01 (1%)
weights_asset_A <- seq(from = 0, to = 1, by = 0.01)

# 3. Create an empty data frame (table) to store all our results
frontier_data <- data.frame(
  Volatility = numeric(),
  Expected_Return = numeric(),
  Correlation = character() # We treat correlation as a category/text for
    coloring the chart
)

# 4. Nested Loops: Calculate stats for EVERY weight, for EVERY
  correlation
for (cor in correlations_to_test) {

  for (weight_A in weights_asset_A) {

    # Since they must sum to 1, Asset B's weight is just 1 minus Asset A's
      weight
    weight_B <- 1 - weight_A

    # Calculate Expected Return
    port_return <- (weight_A * return_asset_A) + (weight_B * return_asset
      _B)

    # Calculate Volatility
    variance_A <- volatility_asset_A^2
    variance_B <- volatility_asset_B^2
    covariance <- volatility_asset_A * volatility_asset_B * cor

    port_variance <- (weight_A^2 * variance_A) +
      (weight_B^2 * variance_B) +
      (2 * weight_A * weight_B * covariance)

    port_volatility <- sqrt(port_variance)
  }
}

```

```

# Save this specific calculation as a temporary row
temp_row <- data.frame(
  Volatility = port_volatility,
  Expected_Return = port_return,
  Correlation = as.character(cor) # Store as text so ggplot colors it
  cleanly
)

# Add this row to the bottom of our main table using rbind (Row Bind)
frontier_data <- rbind(frontier_data, temp_row)
}
}

# 5. Plot the Efficient Frontier using ggplot2
ggplot(data = frontier_data, aes(x = Volatility, y = Expected_Return,
  color = Correlation)) +
  geom_path(linewidth = 1) + # geom_path connects dots exactly in the
  order we calculated them
  labs(
    title = "Two-Asset Efficient Frontier",
    subtitle = "How correlation affects the risk/return tradeoff",
    x = "Portfolio Volatility (Risk)",
    y = "Expected Portfolio Return"
  ) +
  theme_minimal() # A clean, simple background theme for the chart

```

When correlation is +1, the frontier is closest to a straight line and diversification gives little or no curvature benefit. As correlation decreases, the curve bends further left, meaning lower risk becomes possible for similar expected return. When correlation is -1, a special combination can theoretically remove all volatility if the weights exactly offset the fluctuations.

Common Mistake

A very beautiful efficient frontier does not automatically mean the portfolio is realistic. Real markets have transaction costs, liquidity constraints, estimation

error, changing correlations and practical restrictions on short selling.

G. Minimum Variance Portfolio and Sharpe Ratio

Key Definition / Result

The **minimum variance portfolio** is the portfolio with the lowest volatility among the tested weights.

The **Sharpe ratio** is

$$\text{Sharpe Ratio} = \frac{E[R_p] - R_f}{\sigma_p}.$$

It measures extra expected return per unit of volatility. A higher expected return is not automatically better if it requires too much risk.

Suggested AI Prompt: Minimum Variance and Maximum Sharpe Ratio

*** Advanced

"Please extend the two-asset frontier code. For a chosen correlation, calculate the portfolio volatility, expected return and Sharpe ratio for every weight. Identify the minimum variance portfolio and the maximum Sharpe ratio portfolio. Use a 3% risk-free rate."

Investigation / Activity: Comparing Two Kinds of Optimality

Advanced

Teaching Note / Code Solution:

```
# 1. Define our asset characteristics and the risk-free rate
return_asset_A <- 0.12
volatility_asset_A <- 0.20

return_asset_B <- 0.08
volatility_asset_B <- 0.10

risk_free_rate <- 0.03 # 3% Risk-Free Rate
```

```
# Let's choose a correlation of 0 (no relationship) for this analysis
chosen_correlation <- 0

# 2. Create sequences for our weights (from 0 to 1 in steps of 0.01)
weights_asset_A <- seq(from = 0, to = 1, by = 0.01)

# Since weights must sum to 1, Asset B's weight is 1 minus Asset A's
  weight
# Notice we are subtracting an entire list of numbers from 1 all at once
  !
weights_asset_B <- 1 - weights_asset_A

# 3. Calculate metrics for ALL weight combinations simultaneously (
  Vectorization)
# Expected Return
portfolio_returns <- (weights_asset_A * return_asset_A) + (weights_asset
  _B * return_asset_B)

# Volatility
variance_A <- volatility_asset_A^2
variance_B <- volatility_asset_B^2
covariance <- volatility_asset_A * volatility_asset_B * chosen_
  correlation

portfolio_variances <- (weights_asset_A^2 * variance_A) +
  (weights_asset_B^2 * variance_B) +
  (2 * weights_asset_A * weights_asset_B * covariance)

portfolio_volatilities <- sqrt(portfolio_variances)

# 4. Calculate the Sharpe Ratio for every portfolio
# Formula: (Expected Return - Risk-Free Rate) / Volatility
sharpe_ratios <- (portfolio_returns - risk_free_rate) / portfolio_
  volatilities

# 5. Combine everything into a neat master table (Data Frame)
portfolio_data <- data.frame(
  Weight_Asset_A = weights_asset_A,
```

```

Weight_Asset_B = weights_asset_B,
Expected_Return = portfolio_returns,
Volatility = portfolio_volatilities,
Sharpe_Ratio = sharpe_ratios
)

# 6. Identify the Minimum Variance Portfolio (MVP)
# which.min() finds the exact row index where Volatility is the lowest
row_mvp <- which.min(portfolio_data$Volatility)
min_variance_portfolio <- portfolio_data[row_mvp, ] # Extract that
specific row

# 7. Identify the Maximum Sharpe Ratio Portfolio (The Optimal Risky
Portfolio)
# which.max() finds the exact row index where the Sharpe Ratio is the
highest
row_max_sharpe <- which.max(portfolio_data$Sharpe_Ratio)
max_sharpe_portfolio <- portfolio_data[row_max_sharpe, ] # Extract that
specific row

# 8. Print the final discoveries!
cat("\n--- Minimum Variance Portfolio (Lowest Risk) ---\n")
print(min_variance_portfolio)

cat("\n--- Maximum Sharpe Ratio Portfolio (Best Risk-Adjusted Return)
---\n")
print(max_sharpe_portfolio)

```

The minimum variance portfolio and maximum Sharpe ratio portfolio are often different. One asks, “How can I minimise risk?” The other asks, “How can I get the best reward per unit of risk?” This is an important actuarial distinction because different institutions may have different objectives and risk appetites.

Worked Example: Visualising the Two Special Portfolios *** Advanced

Teaching Note / Code Solution:

```
# Make sure the ggplot2 library is loaded
```

```

library(ggplot2)

# Create the visualization step-by-step
frontier_plot <- ggplot(data = portfolio_data, aes(x = Volatility, y =
  Expected_Return)) +

# Layer 1: Draw the Efficient Frontier line
geom_path(linewidth = 1, color = "darkblue") +

# Layer 2: Add a large red dot for the Minimum Variance Portfolio (MVP)
# Notice how we tell ggplot to temporarily use a different dataset for
  this dot!
geom_point(data = min_variance_portfolio,
  aes(x = Volatility, y = Expected_Return),
  color = "red", size = 4) +

# Layer 3: Add a large green dot for the Maximum Sharpe Ratio Portfolio
geom_point(data = max_sharpe_portfolio,
  aes(x = Volatility, y = Expected_Return),
  color = "forestgreen", size = 4) +

# Layer 4: Add text labels next to our dots so we know what they are
# hjust controls horizontal alignment, and we shift the x/y slightly so
  the text doesn't overlap the dot
annotate("text",
  x = min_variance_portfolio$Volatility + 0.003,
  y = min_variance_portfolio$Expected_Return,
  label = "Min Variance (Lowest Risk)", color = "red", hjust = 0,
  fontface = "bold") +

annotate("text",
  x = max_sharpe_portfolio$Volatility,
  y = max_sharpe_portfolio$Expected_Return + 0.002,
  label = "Max Sharpe (Best Trade-off)", color = "forestgreen",
  vjust = 0, fontface = "bold") +

# Layer 5: Add neat titles and axis labels
labs(
  title = "Efficient Frontier with Optimal Portfolios",

```

```

    subtitle = "Highlighting the Minimum Variance and Maximum Sharpe
                points (Correlation = 0)",
    x = "Portfolio Volatility (Risk)",
    y = "Expected Portfolio Return"
  ) +

  # Layer 6: Apply a clean, professional theme
  theme_minimal()

# Display the final plot!
print(frontier_plot)

```

The graph makes the decision trade-off visible. A conservative institution may prefer the red point, while an institution seeking better risk-adjusted return may prefer the green point. Neither point is universally correct without knowing the objective.

H. AI-Assisted Interpretation: Good Prompting and Critical Checking

Synergy & Comparison

AI as a modelling partner: AI can help students write code, explain formulas and suggest visualisations. However, AI output should always be checked against the mathematical meaning. The goal is not to copy code blindly. The goal is to use AI to accelerate exploration while keeping human judgement in control.

Suggested AI Prompt: Asking AI to Interpret Frontier Results

★★

Insight

"I have generated an efficient frontier for two assets in R. The minimum variance portfolio and maximum Sharpe ratio portfolio are different. Please explain in student-friendly language why these two portfolios can differ. Then suggest three reflection questions connecting the result to actuarial risk management."

Investigation / Activity: Critiquing An AI Explanation**** Insight**

Task. Ask an AI tool to explain the difference between minimum variance and maximum Sharpe ratio. Then check whether its answer includes all three of the following ideas:

1. Minimum variance focuses only on reducing volatility.
2. Sharpe ratio balances excess return against volatility.
3. The better portfolio depends on the decision-maker's objective and risk appetite.

Students should learn that AI can give fluent explanations, but fluency is not the same as correctness or completeness. A good actuarial user checks whether the explanation matches the formula and the decision context.

Common Mistake

Do not ask AI only for “the answer”. Ask it to explain assumptions, define variables, comment the code, suggest checks, and identify possible weaknesses. In actuarial work, understanding the assumptions is often as important as producing the calculation.

I. Challenge Extension: When Diversification Fails

Key Definition / Result

Diversification works best when risks are not perfectly positively correlated. It can fail when many assets become exposed to the same hidden factor. In market crises, assets that seemed different may suddenly move together.

Investigation / Activity: Stress Testing Correlation***** Challenge**

Task. Suppose a portfolio was designed under correlation -0.3 , but during a stress period the correlation becomes $+0.8$. Compare the volatility of the same portfolio weight under both assumptions.

Teaching Note / Code Solution:

```
# 1. Define the volatilities (from our previous exercises)
volatility_asset_A <- 0.20
volatility_asset_B <- 0.10

# 2. Choose a portfolio weight to test (Let's use a 50/50 split)
weight_A <- 0.50
weight_B <- 0.50

# 3. Define our two scenarios
correlation_normal <- -0.3
correlation_stress <- 0.80

# 4. Calculate Variance and Volatility for the NORMAL scenario
covariance_normal <- volatility_asset_A * volatility_asset_B *
  correlation_normal

variance_normal <- (weight_A^2 * volatility_asset_A^2) +
  (weight_B^2 * volatility_asset_B^2) +
  (2 * weight_A * weight_B * covariance_normal)

volatility_normal <- sqrt(variance_normal)

# 5. Calculate Variance and Volatility for the STRESS scenario
covariance_stress <- volatility_asset_A * volatility_asset_B *
  correlation_stress

variance_stress <- (weight_A^2 * volatility_asset_A^2) +
  (weight_B^2 * volatility_asset_B^2) +
  (2 * weight_A * weight_B * covariance_stress)

volatility_stress <- sqrt(variance_stress)

# 6. Calculate the percentage increase in risk
```

```

risk_increase_pct <- ((volatility_stress - volatility_normal) /
  volatility_normal) * 100

# 7. Print the results clearly
cat("--- Portfolio Stress Test Results ---\n")
cat("Portfolio Weight: 50% Asset A, 50% Asset B\n\n")

cat("Normal Volatility (Correlation = -0.3):", round(volatility_normal *
  100, 2), "%\n")
cat("Stress Volatility (Correlation = +0.8):", round(volatility_stress *
  100, 2), "%\n\n")

cat("Risk Jump During Crisis: The portfolio volatility increased by",
  round(risk_increase_pct, 2), "%!\n")

```

The same portfolio can become much riskier if correlation increases. This is why risk managers should not only optimise under normal conditions; they should also stress test assumptions.

Reflection Question: The Actuarial Mindset

Why is it dangerous to say, “This portfolio is safe because the historical correlation was low”? What could change in the future?

Session: Real-World 4-Asset Portfolio & Monte Carlo Simulation

In this session, we transition from theoretical 2-asset models to evaluating a 4-asset portfolio using real market data. We will calculate the portfolio's expected return and volatility without using complex matrix algebra, and then deploy a Monte Carlo simulation to uncover the Efficient Frontier.

Task 1: Evaluating a 4-Asset Portfolio Using Real Data

Using the `quantmod` package, we can pull historical stock data and calculate daily returns. By blending these individual returns into a single "Portfolio Daily Return" timeline, we

completely bypass the need for a complex covariance matrix.

Listing 1: 4-Asset Portfolio Evaluation without Matrices

```
# Load the library to fetch financial data
library(quantmod)

# 1. Download real stock data (AAPL, MSFT, GOOG, AMZN)
getSymbols(c("AAPL", "MSFT", "GOOG", "AMZN"), from = "2023-01-01", to = "
2024-01-01")

# 2. Extract Adjusted Close prices and calculate daily returns
# ROC stands for Rate of Change
returns_AAPL <- ROC(Ad(AAPL), type = "discrete", na.pad = FALSE)
returns_MSFT <- ROC(Ad(MSFT), type = "discrete", na.pad = FALSE)
returns_GOOG <- ROC(Ad(GOOG), type = "discrete", na.pad = FALSE)
returns_AMZN <- ROC(Ad(AMZN), type = "discrete", na.pad = FALSE)

# 3. Define equal weights (25% each, must sum to 1.0)
w_AAPL <- 0.25; w_MSFT <- 0.25; w_GOOG <- 0.25; w_AMZN <- 0.25

# 4. Calculate the DAILY portfolio return timeline
port_daily_returns <- (w_AAPL * returns_AAPL) + (w_MSFT * returns_MSFT) +
(w_GOOG * returns_GOOG) + (w_AMZN * returns_AMZN)

# 5. Annualize Expected Return and Volatility (assuming 252 trading days)
annual_return <- mean(port_daily_returns) * 252
annual_vol <- sd(port_daily_returns) * sqrt(252)

# Print results
cat("Expected Annual Return:", round(annual_return * 100, 2), "%\n")
cat("Annualized Volatility (Risk):", round(annual_vol * 100, 2), "%\n")
```

Task 2: Monte Carlo Simulation for the Efficient Frontier

Calculating every exact weight combination for 4 assets is mathematically overwhelming. Instead, we use a Monte Carlo simulation to generate thousands of random weight combinations. The upper-left boundary of the resulting "cloud" of portfolios will naturally reveal the Efficient Frontier.

Listing 2: Monte Carlo Efficient Frontier Simulation

```

library(ggplot2)

# Define simulation parameters
num_portfolios <- 2500
sim_returns <- numeric(num_portfolios)
sim_volatilities <- numeric(num_portfolios)
sim_sharpes <- numeric(num_portfolios)
risk_free_rate <- 0.03

# Run the Monte Carlo Simulation
for (i in 1:num_portfolios) {

  # Generate 4 random numbers and normalize them so they sum to 1 (100%)
  rands <- runif(4, min = 0, max = 1)
  weights <- rands / sum(rands)

  # Calculate daily portfolio returns using the timeline approach
  daily_ret <- (weights[1] * returns_AAPL) + (weights[2] * returns_MSFT) +
    (weights[3] * returns_GOOG) + (weights[4] * returns_AMZN)

  # Annualize and store the risk/return metrics
  ann_ret <- mean(daily_ret, na.rm = TRUE) * 252
  ann_vol <- sd(daily_ret, na.rm = TRUE) * sqrt(252)

  sim_returns[i] <- ann_ret
  sim_volatilities[i] <- ann_vol
  sim_sharpes[i] <- (ann_ret - risk_free_rate) / ann_vol
}

# Combine simulated data into a Data Frame for Plotting
portfolio_cloud <- data.frame(Volatility = sim_volatilities,
                             Expected_Return = sim_returns,
                             Sharpe_Ratio = sim_sharpes)

# Plot the Efficient Frontier Cloud
ggplot(portfolio_cloud, aes(x = Volatility, y = Expected_Return, color = Sharpe_
  Ratio)) +
  geom_point(alpha = 0.5, size = 1.5) +
  scale_color_viridis_c(option = "plasma") +
  labs(title = "4-Asset Efficient Frontier (Monte Carlo Simulation)",

```

```
subtitle = "Simulating 2,500 random tech portfolios",  
x = "Annualized Volatility (Risk)",  
y = "Annualized Expected Return",  
color = "Sharpe Ratio") +  
theme_minimal()
```

J. Consolidation Questions

Investigation / Activity: Review Set A: Risk Register and Scoring ★ Core

1. Write two new risk statements using the structure “If ..., then ...”.
2. Assign likelihood and impact scores from 1 to 3.
3. Explain why the risk score is useful but imperfect.
4. Identify one treatment for each risk.
5. Explain the difference between inherent risk and residual risk.

Investigation / Activity: Review Set B: Frequency and Severity ★★ Intermediate

1. Define frequency and severity in the claim simulation.
2. Explain why total loss is random.
3. Interpret the mean, standard deviation and 95th percentile of simulated total losses.
4. **Explain why a Uniform distribution might not be realistic for insurance severity (e.g., real claims have no strict upper limit and tend to be skewed).**
5. Describe one limitation of our Binomial frequency model.

Investigation / Activity: Review Set C: Correlation and Portfolio Risk***** Advanced**

1. State the two-asset portfolio variance formula.
2. Identify the covariance or correlation term in the formula.
3. Explain why lower correlation can reduce portfolio volatility.
4. Compare the meaning of minimum variance and maximum Sharpe ratio.
5. Explain how stress testing correlation helps risk management.

Investigation / Activity: Challenge Questions***** Challenge**

1. If two risks have correlation zero, are they necessarily independent? Explain.
2. Can diversification eliminate all risk in real financial markets? Why or why not?
3. Why might a low-return asset still appear in a good portfolio?
4. How could model error affect an efficient frontier?
5. How would transaction costs, liquidity risk or short-selling constraints change the decision?

K. Wrap-Up: The Full Story

Key Definition / Result**Today's big picture**

1. A risk register can become structured data.
2. Frequency and severity together shape total loss.
3. Simulation helps us explore uncertainty when formulas are difficult.

4. PMFs, PDFs, means and standard deviations summarise random variables.
5. Covariance and correlation explain how risks interact.
6. Portfolio variance shows why diversification can work.
7. The efficient frontier helps compare risk-return choices.
8. Minimum variance and maximum Sharpe ratio represent different decision goals.
9. Stress testing reminds us that assumptions may change.

Synergy & Comparison

Final Actuarial Message: Risk management is not the art of avoiding every uncertain outcome. It is the discipline of understanding objectives, modelling uncertainty, checking dependence, comparing trade-offs, and making transparent decisions. R and AI are powerful tools, but the actuary's judgement remains the centre of the process.

End of Tutorial 6